Concepção e Autoria Multimédia

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Methodologies for Multimedia Project Development

- Methodologies for Multimedia Project Development have, in most instances, focused on **what is being developed** with less attention to how the product is developed.
- Until recently, the multimedia environment had few guidelines so that **development activities and methods varied** according to the background of the producer, for example, publishing, computing or video production.
- However, while varied, these models for design and production have in common **activities which are team based and complex**, they are all producing a particular end product as a result of the team effort and each of these products has unique properties.

Methodologies for Multimedia Project Development

- Multimedia information systems: databases, information kiosks, hypertexts, electronic books, and multimedia expert systems
- Multimedia communication systems: computer supported collaborative work, videoconferencing, streaming media, and multimedia teleservices
- Multimedia entertainment systems: 3D computer games, multiplayer network games, infotainment, and interactive audio-visual productions
- Multimedia business systems: immersive electronic commerce, marketing, multimedia presentations, video brochures, virtual shopping…
- Multimedia educational systems: electronic books, flexible teaching materials, simulation systems, automatic testing, distance learning…

Methodologies for Multimedia Project Development

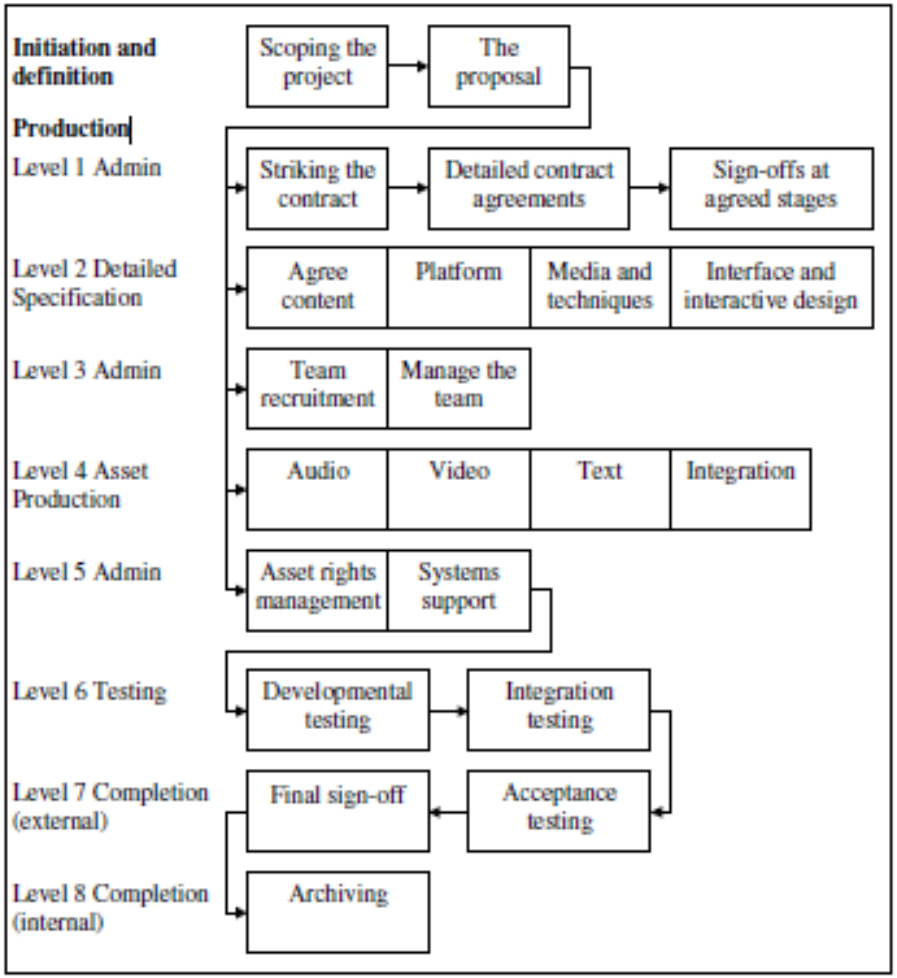


- Initiation
- Definition
- Specification and design
- Development
- Delivery
- Evaluation.
- Project management is a continuing and integral part of these development phases.
- Evaluation and testing are also ongoing phases, and include evaluation and testing of both the product being developed and the process being followed.

Structured Project Development

Initiation	Specification	Design	Execution	Controlling	Closure
Validating Needs	Project Scoping	Technical Specification	Assets Preparation	In-House Testing	Client Testing
Collecting Data	Collecting Clients Requirements	Conceptual (Functional) Design	Scripting and Programming	Correction and Review	Market Testing
Choosing Project Management Team	Risk Evaluation	Final Project Plan	Scripting and Programming Review		Correction and Review 2
Feasibility Study	Project Breakdown	Training	Assembling		Preparing and Signing Acceptance Document
	Task Identification	Prototyping	Follow Up		
	Primary Schedule Preparation	Prototype Review			
	Proposal & Acceptance	Structural (Technical) Design			
	Contract Preparation and Signing				
	Choosing Team				

Structured Project Development (Hannington & Reed)

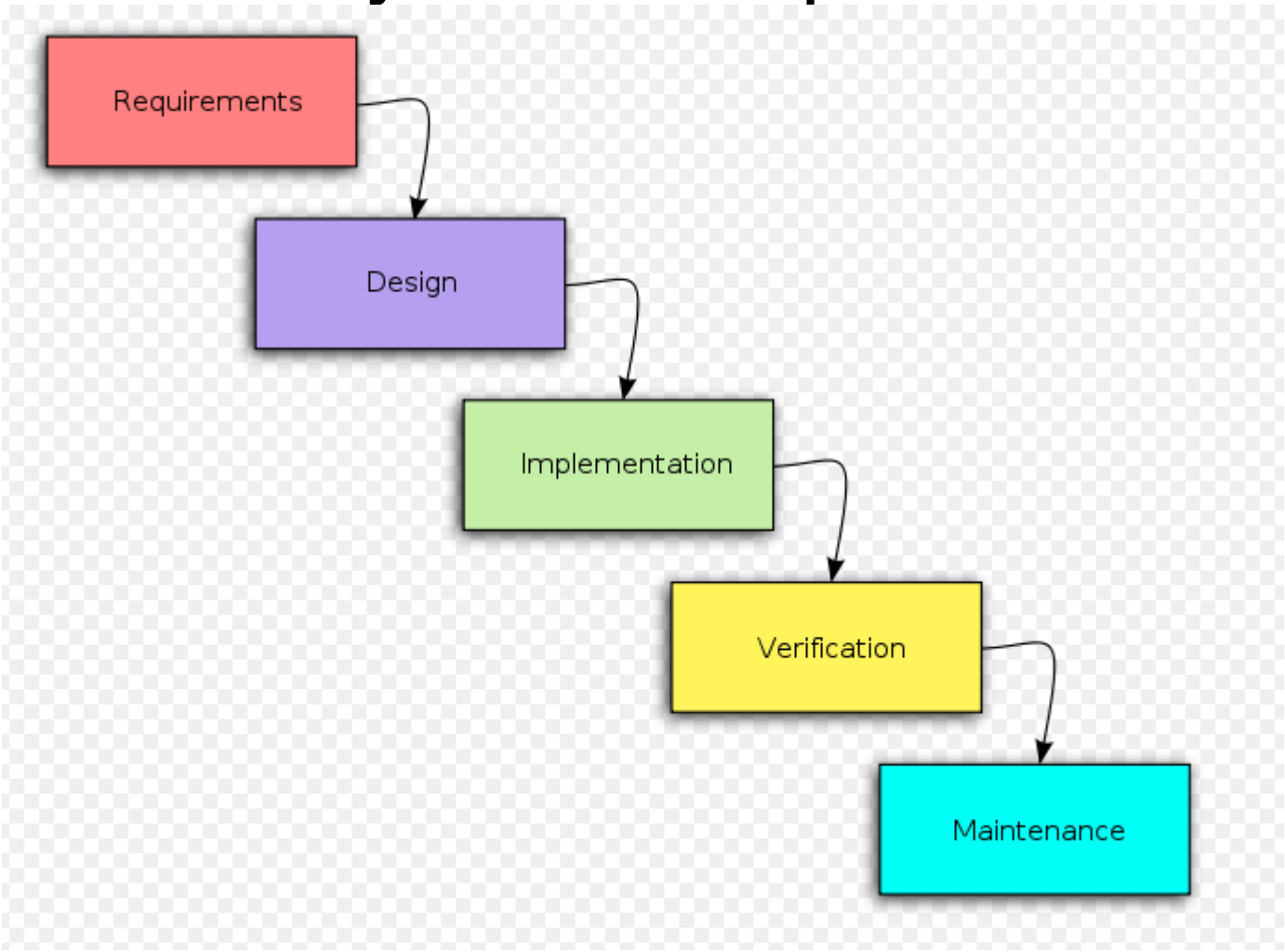


OOHDM - Object-Oriented Hypermedia Design Method

Activities	Products	Formalisms	Mechanisms	Design Concerns
Requirements Gathering	Use Cases, Annotations	Scenarios; User Interaction Diagrams; Design Patterns	Scenario and Use Case Analysis, Interviews, UID mapping to Conceptual Model	Capture the stakeholder requirements for the application.
Conceptual Design	Classes, sub-systems, relationships, attribute perspectives	Object-Oriented Modeling constructs; Design Patterns	Classification, aggregation, generalization and specialization	Model the semantics of the application domain
Navigational Design	Nodes, links, access structures, navigational contexts, navigational transformations	Object-Oriented Views; Object-Oriented State charts; Context Classes; Design Patterns; User Centered Scenarios	Classification, Aggregation, generalization and specialization.	Takes into account user profile and task. Emphasis on cognitive aspects. Build the navigational structure of the application
Abstract Interface Design	Abstract interface objects, responses to external events, interface transformations	Abstract Data Views; Configuration Diagrams; ADV-Charts; Design Patterns	Mapping between navigation and perceptible objects	Model perceptible objects, implementing chosen metaphors. Describe interface for navigational objects. Define lay-out of interface objects
Implementation	Running application	Those supported by the target environment	Those provided by the target environment	Performance, completeness



Structured Project Development



Rapid Prototyping

- Customer satisfaction by **rapid, continuous delivery of useful software**
- **Working software is delivered frequently** (weeks rather than months)
- **Working software is the principal measure of progress**
- Even **late changes in requirements are welcomed**
- Close, **daily cooperation** between business people and developers
- **Face-to-face conversation** is the best form of communication (Co-location)
- Continuous attention to **technical excellence and good design**
- **Simplicity**
- **Self-organizing teams**
- Regular **adaptation to changing circumstances**

Effective Use

Approach	*General Industry (n = 8)*		*Multimedia Industry (n = 15)*		*Aggregate Response (n = 23)*	
	Incidence		Incidence		Incidence	
Semi-structured SDLC	8	100%	6	40%	14	61%
Prototyping	2	25%	9	60%	11	48%
Production-oriented Approach	1	13%	8	53%	9	39%
Structured SDLC	3	38%	4	27%	7	30%
Advertising / Graphic Design	2	25%	5	33%	7	30%
Object-oriented Approach	3	38%	3	20%	6	26%
Other	1	13%	4	27%	5	22%
Artistic Approach	0	0%	4	27%	4	17%
Media Design Approach	0	0%	4	27%	4	17%

Effective

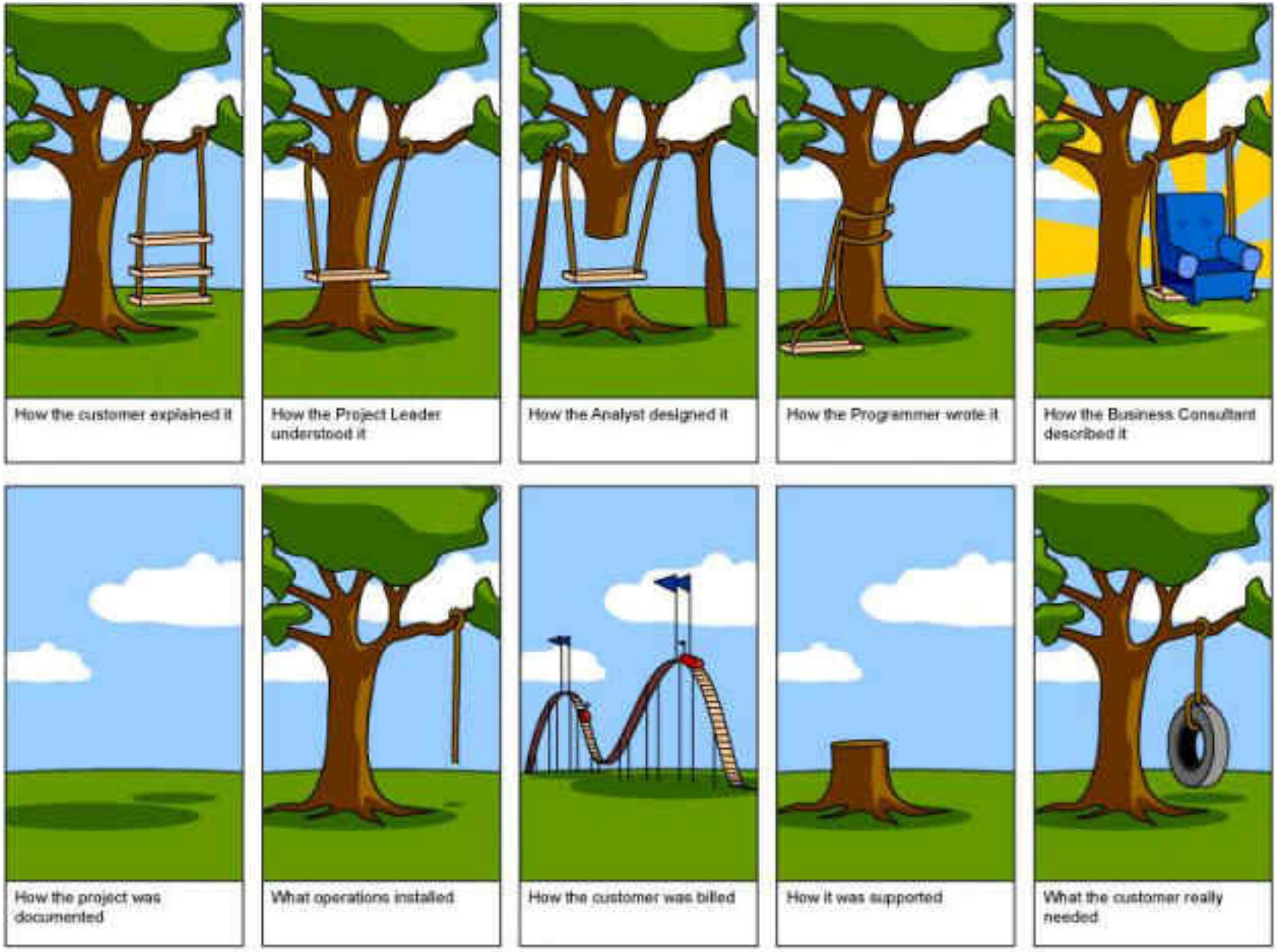
Technique	*General Industry (n = 7)*	*Multimedia Industry (n = 15)*	*Aggregate Response (n = 22)*	
	(Affirmative responses)	(Affirmative responses)	(Affirmative responses)	
Project Management	6	13	19	86%
Prototyping	5	11	16	73%
Flowcharting	3	12	15	68%
Storyboarding	3	10	13	59%
Menu Maps	2	9	11	50%
Data Flow Diagrams (DFD)	2	7	9	41%
Object-Oriented techniques	2	2	4	18%
Relationship Management Data-Model (RMDM) Diagram	1	3	4	18%
Movie Authoring and Design (MAD)	1	3	4	18%
Class Diagrams	2	1	3	14%
Entity Relationship Diagrams (ERD)	1	2	3	14%
Dialogue Charts	0	3	3	14%
Other	1	1	2	9%
State Transition Diagrams (STD)	0	2	2	9%
Functional Decomposition Diagrams (FDD)	1	0	1	5%
Use Case Diagrams	1	0	1	5%
Joint Application Design (JAD)	0	1	1	5%

Methodology



Project Development Phases

- Initiation
  - Problem identification
    - You have a problem that you believe may benefit from a multimedia solution. Through close examination, you determine the exact nature of the problem. You may want to document this in a concise problem statement.
    - You research a range of potential solutions (multimedia and others). You collate a shortlist of potential solutions and possibly revise the problem statement.
    - Now, you can analyze the relative merits of the solutions by considering factors such as technical feasibility, existing skills, and existing infrastructure.
  - You can then select a potential solution and develop a project brief that documents all information gathered about the problem, the potential solution, budget and schedule parameters.


How the customer explained it
How the Project Leader understood it
How the Analyst designed it
How the Programmer wrote it
How the Business Consultant described it
How the project was documented
What operations installed
How the customer was billed
How it was supported
What the customer really needed

Project Development Phases

- Initiation
  - Initial Team Selection
    - You have a documented understanding of the problem and the potential multimedia solution. You decide whether you can develop that potential solution yourself or whether you will need to outsource it.
    - If you decide to outsource the project you will now approach a number of service providers to request a proposal or scope document in response to your project brief. Based on the responses from the service providers you will select the service provider with whom you will negotiate a binding project contract.
    - If you decide to develop the package yourself you will now select the initial members of the project team.


Gerente de Logistica
Gerente de Marketing
Gerente de Recursos Humanos
Gerente de Segurança
Gerente de Recursos Humanos
Gerente de Comunicação
Gerente do Projeto
Supervisor Interno
Gerente de Produção
Gerente de Desenvolvimento de Produtos
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Project Development Phases

- Initiation
  - Initial Team Selection
    - Some of the team members typically involved are:
      - instructional/information designers
      - graphic designers
      - programmers/software developers
      - project managers
      - production managers
      - media producers
      - external testing and evaluation houses.
    - The actual make-up of the team depends on the nature of the product to be developed and the skills base of the team.
    - It may be that not all team members are identified up-front, but rather that they join the team as required.
    - The team members selected at this stage will probably be the project manager and possibly the multimedia designer.

Project Development Phases

- Definition
  - The purpose of this phase is to determine what is to be built, by whom and how.
  - Needs analysis
  - Scope document
  - Project contract
  - Team selection
  - Project plan

Project Development Phases

- Definition – Needs analysis
  - While completing the **needs analysis** you will gather detailed information on all aspects of the project, including:
    - target audience
    - content
    - a technology profile of the target environment
    - technology needs for the development
    - budget and schedule
    - staff requirements
    - risks.

Project Development Phases

- Definition
  - A **scope document** (also called a concept brief or project proposal) is then developed as the project team's detailed proposal for meeting the project needs.
  - The next step is to negotiate a **project contract**. Having secured a contract (or the go-ahead if it is an in-house project), the selection of the project team then proceeds.
  - A **project plan** may also be developed at this stage. The project plan describes and guides the project management activities that will be necessary for the effective execution and control of the project.

Project Development Phases

- Definition – Needs Analysis
  - Identify the Target Audience
  - Determine the Project Scope
  - Identify Software and Hardware Requirements
  - Project and Budget Schedule
  - Human Resources
  - Project Feasibility Study
  - Various techniques (such as **interviews with the client and questionnaires** with samples of the target audience) can be used to define the project needs.
  - The process of needs analysis may result in a separate needs report, especially on large projects. On smaller projects, the needs analysis and the information gathered can often be documented with the proposed solution in the one document, the scope document. This document provides information on which all design decisions will be based in the next stage of development.

Project Development Phases

- Definition – Needs analysis
  - Identify the **Target Audience**
    - It is essential to determine the needs of your target audience before beginning the design process more formally, as different audiences have different needs and abilities.
    - Consider the following:
      - the level of education of the users
      - the background knowledge and skills of the users
      - degree of experience and skill with technology
      - attitudes and expectations
      - age
      - access to equipment
      - physical location
      - cultural factors
      - the length of time the users will be using the package
  - Consider the needs of the widest range of users including people with disabilities, slow connections, early browser versions, etc. Designing for the widest range of users can also benefit users without disabilities or technical limitations.

Project Development Phases

- Definition - Needs analysis
  - Determine the **Project Scope**
    - Determine the **boundaries of the content** of the package, in breadth and depth, as well as for the graphical treatment.
    - The scope of the content may change through the development as both you and the project team develop a clearer idea of what the package will contain and look like. (The secret of effective scope management lies in determining how changes will be made, and deciding until when in the project process changes to the content scope will be made).

Project Development Phases

- Definition – Needs Analysis
  - Determine the Project Scope
    - The following questions will need to be considered. These will help crystallize the nature and scope of the product.
      - What is the scope of the content? What concepts will be in the package? What concepts will not be in the package?
      - What combination of media will be used to present it?
      - Is there a metaphor that will be adopted for the whole package or part of the package?
      - Are there any particular requirements for the design treatment of the media elements?
      - How long do I expect the user to sit at the computer in any user session?
      - Does the material need to be presented linearly or is it more appropriate for users to 'browse'?
      - Should the user only be allowed to access some sections after completing certain activities in other sections?
      - Who will design, develop and produce the media?
      - Who will review and evaluate the media?
      - Who will authorize changes to the content scope?
      - Until what stage in the process will changes to the content scope be necessary or/and allowed?

Project Development Phases

- Definition – Needs analysis
  - Identify Software and Hardware Requirements
    - In order to develop a multimedia product which can be accessed by all target users, the project development team will need to know the following information:
      - What basic hardware (CPU, monitor, RAM, CD-ROM speeds, modem) do the target users have?
      - What basic software (operating system, browser, plug-ins, and drivers) do the users have?
      - In what environment (e.g. library, office, home) will the users access the package?
      - What hardware is available to and required by the development team?
      - What software is available to and required by the development team?
      - What are the storage requirements?
      - What are the security requirements?
      - What user management systems will need to be in place?
      - What will be the minimum infrastructure support and maintenance requirements (e.g. server downtime)?

Project Development Phases

- Definition – Needs analysis
  - Identify Software and Hardware Requirements
    - It may be that more powerful machines (machines on which to develop complex animations, for example) need to be purchased for the development of this product.
    - It will be necessary to survey existing software and to assess its appropriateness as a development tool.
    - It may be necessary to purchase additional software to complete the development.
    - It would also be wise to review potential changes that may occur in hardware and software during the development cycle.

Project Development Phases

- Definition– Needs analysis
  - Project and Budget Schedule
    - You need to know what funding is available for the project (or at least have a general idea) so that you can decide whether what is required can be designed and developed within that budget. It may be necessary to outline a few alternative solutions or development approaches with varying costs, to provide your client/funding body with an understanding of the cost, quality and schedule relationship.
    - You may be able to save yourself a significant amount of time by having a clear idea of the available budget. Many hours can be wasted developing a scope document that is rejected because it is outside an achievable budget.
    - As with the budget, you need to know what the expectations for the package are, so that you can work out whether what is required can be developed within this timeframe. There is often a significant gap between a client's and the project team's idea of how much time is needed for developing a package. The developing team may use past project experience to develop schedule estimates and demonstrate to the client that the team's views are reasonable.
    - Schedule and budget overruns are typical risks that face multimedia projects. Overruns can be caused by underestimating the time needed, both to write the content (by the client) and to develop the multimedia (by the development team)

Project Development Phases

- Definition – Needs analysis
  - Human Resources
    - From the needs identified so far and the project manager's experience, the skills and staff required for the project can be determined. It is necessary to have a good idea of what staff may need to be recruited and how this may impact on the budget and schedule.
    - If you have chosen to outsource the development of your product, the human resources will be the responsibility of the organisation you are dealing with. You will need to identify the content expert, if it is someone other than yourself, and organise their participation in team meetings.

Project Development Phases

- Definition – Needs analysis
  - Project Feasibility Study
    - As part of the needs analysis, the feasibility of the project, including a cost/benefit analysis, should be determined. Basically, this is a process of establishing whether the wants and needs can be achieved within the project constraints (budget, schedule, personnel, and skills). How extensive and critical the feasibility study is depends on the size and nature of the project. You may wish to undertake an initial risk assessment.
    - A cost/benefit analysis of the potential solutions may also be done to guide the selection and support the proffered solution



Project Development Phases

- Definition – Scope document
  - The aim of the scope document is to document and justify the proposed solution. (This document is sometimes called a concept brief or project proposal). It should contain most, if not all, of the information that will form the project contract. Information gathered in the needs analysis is also documented here, even if on many projects this may already be documented as a needs report.
  - Client and Objectives
  - Project Product
  - Resources
  - Intellectual Property
  - Limitations of the Document

Project Development Phases

- Definition - Scope Document - Client and Objectives
  - You will need to identify the following:
    - Project details:
    - the full name of the project and any acronyms or abbreviated name
    - sponsor/client
    - project developer
    - target audience (demographic details of the target audience as ascertained by the needs analysis)
    - other relevant authorities and contacts
    - name of the organisation
    - Address, telephone numbers and e-mail addresses.
    - Project background:
    - a brief history of how and who originated the project and its development to date
    - general project objectives
    - other client objectives
    - other team objectives
    - target audience needs
    - other benefits to stakeholders.

Project Development Phases

- Definition - Scope Document - Project Product
  - Deliverables - All multimedia and documentation deliverables. (This will be a statement about the number and format of all products that are included in the project – CD-ROMs, Web sites, accompanying booklets, videos, etc.)
  - Treatment - Description and justification of the proposed treatment - media, delivery format, style, including possible variations.
  - Proposed structure and navigation - Outline diagram of the proposed structure and navigation.
  - Performance requirements - The performance requirements (minimum and desirable) of all parts of the product.
  - Delivery requirements - The requirements related to final delivery, packaging, final product technical and user documentation.
  - Implementation environment - The technological profile of the target environment and its implications on the performance requirements for hardware and software.

Project Development Phases

- Definition - Scope Document
  - Resources - The following are some elements which will need to be included in the scope document.
    - Content and source - The scope of the content, the source, what will be coming from the client and when, the media elements and source of production.
    - Budget and payment structure - Proposed budget and the payment schedule.
    - Schedule, work breakdown and milestones - Proposed schedule, and general breakdown of the tasks, relevant milestones referring to the proposed lifecycle.

Project Development Phases

- Definition - Scope Document
  - Intellectual Property - This section covers issues such as the ownership of original content, ownership of final product(s), and responsibility for obtaining copyright permission to use copyrighted material owned by third parties.
    - Original content and final product - At all stages, the issue of "Who owns the intellectual property?" of each component (from sketches to final product) will need to be very clear to all parties, the development team, the individual team members and the client. The Intellectual Property (IP) of all intermediate and final products should be specifically addressed in the project contract, and the project plan should define the techniques and activities to be used to monitor and document IP throughout the development of the project. Make sure there is a reference in the project plan to this.
    - During the development of the project you may find it necessary to define and confirm IP ownership. This is especially the case if contract media producers are engaged or if the team brings in products and tools from previous work. You might like to keep an IP register to log the IP ownership of each resource brought into, used or produced by the project.

Project Development Phases

- Definition - Scope Document
  - Intellectual Property
    - Third party materials - The intellectual property or copyrights of third parties also have to be respected. This applies to all copyright materials, including those accessed from the Web. The specific needs and responsibility for obtaining permission for use of copyright material will have been defined and agreed to by stakeholders in the project plan and project contract.

Project Development Phases

- Definition - Scope Document
  - Limitations of the Document - because the understanding of the needs and the final solution of the project may change through the iterative processes of development, you may wish to include a statement on the limitations of the document. This would probably include:
    - the purpose of the document
    - how and by whom it can be used
    - the time period to which it applies
    - assumptions based on client information
    - factors to consider - including issues which may be still under discussion but not yet quoted and scoped for, and features, functionality or content which have been specifically excluded in this iteration
    - future options - including features and functionality which have been discussed, but not quoted or scoped for in this version of the product.



Project Development Phases

- Definition - Project Contract
  - The reality of multimedia development is that the first draft of the scope document is rarely fully mutually agreed upon. There are usually negotiations to refine the specifications of the deliverables. These will of course impact on the budget and schedule.
  - The final scope document should clearly specify the milestones and sign-off points, including possible points and conditions for revisions to the budget and schedules. A timeframe should be included in the document, but a full timeline may not necessarily be part of the document at this stage.
  - As part of the project contract, there must be clear agreement on issues such as reporting and communication, documentation, evaluation and testing and delivery requirements. This defines how you and the project team will work together and how, through the agreed process of sign-offs and revisions, a mutual agreement will be reached that the right product has been built in the agreed way.

Project Development Phases

- Definition - Project Contract
  - The actual signing of the contract may involve authorised representatives signing a specified agreement on the last page of the scope document, with initials of both parties on each page. Any subsequent reviews or other documentation (e.g. design document, evaluation reports, budget and schedule reviews) which varies this agreement, will also have to be signed by the authorised representatives.
  - Throughout the project, you and the development team have a mutual obligation to inform each other of any event that may impact on successful progress and timely completion of the project.

Project Development Phases

- Definition - Team Selection
  - By this stage, you will already have a project manager selected for your team. Selection of the remaining team members should occur under the supervision of the project manager.
  - In the Initiation phase you will already have completed an assessment of the skills required by the development team. You may also have completed a skills audit of your staff to match available skills against required skills.
  - You may need to revisit that analysis now to verify that it is correct and to ensure that the staff identified are committed to completing the project and have time to do so. The development of timelines and milestones will be dependent upon the availability of team members. It is crucial that the long-term availability of all team members be assessed and compared.

Project Development Phases

- Definition - Team Selection
  - It may be necessary for team members to be provided with release time in order to meet their commitments for the project.
  - There may also be the need for secretarial or data entry support for the team. In some cases, team members may need technical help to produce content in a format useable by the development team, for example, assistance with scanning or digitising media.

Project Development Phases

- Definition - Team Selection
  - Development Team Roles
    - Project Manager - the role of the project manager includes:
      - client liaison
      - content expert liaison
      - preparation of agreement, sign-offs and timelines
      - managing project teams
      - keeping project on time and on budget
      - keeping track of all project documentation
      - managing the external testing procedures if required
      - managing the replication process.
    - Production Manager - the role of the production manager includes:
      - assisting project teams in technical design issues
      - providing guidance regarding programming approaches to be taken
      - assisting in team selection for each job
      - supervision of project teams
      - assisting with programming documentation
      - programming (see tasks listed under Programmers/Software designers, below)
      - assisting in file management issues.

Project Development Phases

- Definition - Team Selection
  - Development Team Roles
    - Content experts - responsible for:
      - scoping/writing content
      - identifying/describing/producing/finding media needs
      - obtaining copyright clearance for content produced by others
      - verifying accuracy of content
      - editing and proofreading content before and after implementation.
    - Graphic designers - the role of the graphic designers includes:
      - design and development of major interface artwork
      - design and development of global interface items (e.g. buttons, labels, scrollbars, dialogue boxes)
      - design of additional interface items (such as title screen, credits screens, help screens)
      - preparing graphic design specifications documentation
      - development of major interface artwork
      - development of global interface items and additional interface items
      - preparing CD-face artwork, insert artwork (the booklet in the front of a CD case) and inlay artwork (at the bottom of a CD case)
      - image scanning and manipulation
      - creation of content illustrations, diagrams and figures
      - creation of animations (3-D or 2-D, complex or simple)
      - testing/debugging process (graphics Quality Assurance (QA))

Project Development Phases

- Definition - Team Selection
  - Development Team Roles
    - Programmers/Software designers - the role of the programmers/software developers includes the following:
      - program analysis and design
      - algorithm selection, adaptation and design
      - data structure and database design
      - design of fine-grained interface behaviour
      - preparation of technical specifications.
      - programming (Actionscript, Visual Basic, C++, MS Access, CGI, ASP, etc.)
      - code documentation
      - incorporation of all interface graphics
      - incorporation of all content (including diagrams, animations, text, video, sound, images, etc.) incremental testing of code modules
      - gold master CD replication.
      - internal testing, debugging

Project Development Phases

- Definition - Team Selection
  - Development Team Roles
    - Other team members - the team can also include:
      - a video production unit
      - a video digitising unit
      - a sound production unit
      - voice or video talent
      - data entry staff
      - a software testing house

Project Development Phases

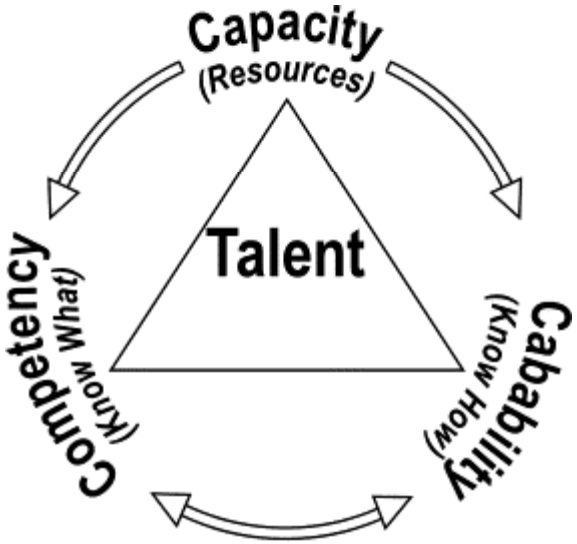
- Definition - Project Plan
  - A project plan (or project management plan) guides the execution and control of the project. In your organisation there may already be well documented and/or established ways of executing and controlling projects, so that no separate project plan needs to be generated. In such cases, relevant sections of the scope document may be detailed enough to guide the development of the specific project.
  - However, especially on larger projects, those sections of the scope document may only outline the intentions and processes. A specific project plan may be developed by the project team to flesh out exactly what, how, who, and when project activities are to be done. A plan may be a single detailed document or may be a summary, which is accompanied by several appendices or sub-plans.

Project Development Phases

- Definition - Project Plan
  - A properly managed project will have established processes for controlling and executing the project. Departments or organisations may already have established processes which should be followed, or which may require some project-specific modification. Changes made to the established management process should be documented and circulated to the whole team. Projects that do not have adequate project management rarely meet their time, budget and quality expectations

Project Development Phases

- Definition - Team Selection



Project Development Phases

Definition - Team Selection

- The organisational leadership must be involved and networked. This can be useful if the development stalls due to a resource problem, as the network can be exploited to obtain short-term resource solutions.
- The project manager needs to have technical skills and knowledge of the technology involved as well as being a capable manager. The technical skill requirement has become lost in the majority of literature on product development.
- The project team must have previously worked on like projects. It would be imagined that this would be an obvious point, however, having a team of champions does not necessarily result in a champion team. Severe stress was evident in teams coming together for the first time and working on a project, which was a new type of development for several members.

Project Development Phases

- Team Selection - Creative climate
  - Building a creative climate involves systematic development of appropriate organisational structures, communication policies and procedures, reward and recognition systems, training policy, accounting and measurement systems and deployment of strategy. The problem with the notion of a creative climate, or even individual creativity, is that the term is ambiguous. Creativity will take on a unique definition for each individual trying to implement it. In the management context then, the term creativity is unbounded and hence, may not be able to be managed in its strict sense.
  - Creativity can be described in terms of the dimensions of competency and capability for a given capacity, and each of these dimensions is manageable. Each dimension may be independently resourced to accommodate differing phases and conditions within the project process. The management issue then becomes one of establishing the best starting condition for a team to undertake a product development project.

Project Development Phases

- Team Selection - Project manager
  - In most cases the project manager was drawn from within the host organisation. In this situation, project success directly related to the degree to which the project manager was allowed to focus on the given project. In several instances, the project manager assumed a part-time role on a number of concurrent projects. Hence, the project manager simply collected data for the time charts and budget, and reacted to situations rather than proactively managing the project. **In all instances where part-time project managers were involved, the projects were delivered late.**
  - In order to overcome the project management resource constraint, some organisations opted to outsource the role to a professional project manager. Where this person had considerable technical knowledge of the technologies to be incorporated in the product and with excellent project management experience and able to establish good rapport with the team, the project was completed successfully.
  - In two instances, the individual outsourced project manager was highly experienced in managing projects but lacked an understanding of the technologies involved. In both cases, the developments ran into trouble in the later stages of the projects, due to items that should have been part of the critical path, but everyone assumed someone else was taking responsibility for informing the project manager of its importance. No one had thought about these items until they surfaced, and none had raised them as issues when the planning stages were operating.
  - One of the important characteristics of a successful project manager was the high degree to which they were networked outside the organisation. This proved important where internal resources, either technical or personnel, might not have been sufficient to deal with an emerging problem. In this situation and with the "networked" project manager, help tended to be "just a call away".

Project Development Phases

- Team Selection - Leadership
  - Two leadership roles were identified from this research; organisational, with general leadership responsibility and functional with specific technical areas of responsibility. In the larger organisations involved in the research project, it was usual for a senior manager (CEO) to be responsible for the organisation and have no detailed involvement or even passing interest in the particular product development project. In this instance it was a functional manager (say Project Manager) who took primary responsibility for the project on behalf of the organisation. The importance of the organisation and functional leaderships roles is discussed below.
  - In the case of smaller organisations, the division of leadership roles became increasingly merged as the organisation became smaller. In some instances, organisational or functional leadership delegated the entire project to the project manager. Hence, the project manager lost focus, due to spending many hours with stakeholders working on peripheral issues. In this case the project managers became part-time managers, with the outcome associated with this approach to project management.

Project Development Phases

Team Selection – Teams

- Team structural composition varied from fully outsourced, composite outsourced-insourced but assembled in-house, in-house teams that had worked together for a long period of time and newly assembled teams from in-house resources. For a team to be successful it must be productive. Where teams had a comfortable working relationship between all players and a definite team culture, it was expected and confirmed that such teams were effective and productive.
- However, the major problem related to a predetermined approach to problem solving. Although such teams were in a development environment associated with innovation, they had become specialists at a particular approach and not "innovative about solutions". It seemed likely that more effective and efficient approaches were not considered. In a sense this would appear a reasonable approach to limiting risk associated with a new approach, and this would be reasonable in a static development environment. However, the environment is not static and skill redundancy is a major issue. Without an innovative approach to innovation, skill redundancy has a high probability of occurrence with individual members of an established team.

Project Development Phases

- Team Selection - Teams
  - The advantage of outsourced members came from a "cross-fertilisation" of ideas and approach to the project task. In this type of team the human resource role of the project manager proved essential in the management of composite teams.
  - Team structure itself affected the resource management task of the project manager. Three important factors, human resource (HR) skill, time management skill and technical skill were identified as specific competency components of the project manager in addition to the competencies expected for application to a project. The three skill dimensions were required to be applied differently dependent on the structure of a particular team.
  - The worst possible team environment occurred when there was a combination of an ineffective project manager and non-resident team members. The "tyranny of distance" reinforced negative aspects associated with communication barriers and led to project breakdown.

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